Hoff Bayesian Statistics

Contents

These are (fully reproducible!) R Markdown lecture notes for Peter D. Hoff, "A First Course in Bayesian Statistical Methods"[1], completed as part of a 1-semester independent study course. Only Chapters 1-8 are complete right now.

Each note includes summaries of chapter sections, with math and explanations modified to better fit my understanding and the occasional link to external resources. I also reproduce many figures in the book in a ggplot/tidyverse style, and tackle some of the exercises at the end of each chapter (correctness not guaranteed).

If you find an error or would like to improve the notes, please let me know/submit a PR!

1 Chapters

- Chapter 1: Introduction and examples[2]
- Chapter 2: Belief, probability, and exchangeability[3]
- Chapter 3: One-parameter models[4]
- Chapter 4: Monte Carlo approximation[5]
- Chapter 5: The Normal Model[6]
- Chapter 6: Posterior approximation with the Gibbs sampler[7]
- Chapter 7: The multivariate normal model[8]
- Chapter 8: Group comparisons and hierarchical modeling[9]
- Chapter 9: Linear regression[10]
- Chapter 10: Nonconjugate priors and Metropolis-Hastings algorithms[11]

2 Final Project

As a small final project, I also implemented R code for the basic binary relation version of the Infinite Relational Model, described in Kemp et al. (2006), "Learning Systems of Concepts with an Infinite Relational Model"[12].

- Infinite Relational Model[13]

[1] https://peterhoff.io/book/
[2] 1.qmd
[3] 2.qmd
[4] 3.qmd
[5] 4.qmd
[6] 5.qmd
[7] 6.qmd
[8] 7.qmd
[9] 8.qmd
[10] 9.qmd
[11] 10.qmd
[12] http://web.mit.edu/cocosci/Papers/Kemp-etal-AAAI06.pdf
[13] irm.qmd